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# Initialize and Define Test Matrix

```
clear; clc; close all;
% Test matrix and number of elements to keep
x = [-1 3 4 7; 8 -2 1 1; 0 -13 1 4]
% Find the M largest values by magnitude
M = 5
```

```
x =

    -1     3     4     7
     8    -2     1     1
     0   -13     1     4


M =

     5
```

# Find the Largest Elements

```
xvec = x(:)
% Set the remaining values to zero and reshape the matrix
[~, order] = sort(abs(xvec), 'descend')
selected = sort(order(1:M))
```

```
xvec =

    -1
     8
     0
     3
    -2
   -13
     4
     1
     1
     7
```

```
     1
     4


order =

     6
     2
    10
     7
    12
     4
     5
     1
     8
     9
    11
     3


selected =

     2
     6
     7
    10
    12
```

# Find the row and column positions of selected values

```
yvec = zeros(size(xvec))
yvec(selected) = xvec(selected)


yvec =

     0
     0
     0
     0
     0
     0
     0
     0
     0
     0
     0
     0
```

```
yvec =

     0
     8
     0
     0
     0
   -13
     4
     0
     0
     7
     0
     4
```

# Find Indices and Calculate Energy

```
y = reshape(yvec, size(x));
% Calculate the energy ratio after compression
[K, L] = ind2sub(size(x), selected);
K =K(:)
L =L(:)
if sum(abs(xvec).^2) == 0
    r = 0
else
    r = sum(abs(yvec).^2) / sum(abs(xvec).^2)
end
```

```
K =

     2
     3
     1
     1
     3


L =

     1
     2
     3
     4
     4


r =

    0.9486
```

# Display Results and Verify Function

```
disp('Original matrix:')
disp(x)
% Compare the results with the compress2 function
disp('Compressed matrix:')
disp(y)
disp('Selected positions [K L]:')
```

```
Original matrix:
    -1     3     4     7
     8    -2     1     1
     0   -13     1     4

Compressed matrix:
     0     0     4     7
     8     0     0     0
     0   -13     0     4

Selected positions [K L]:
```

# Plot Original and Compressed Matrices

```
disp([K L])
fprintf('Energy ratio: %.6f\n',r)
[Y2, K2, L2, R2] = compress2(x, M)
assert(isequal(y, Y2) && isequal(K, K2) && isequal(L, L2) && abs(r-R2) <
1e-12);
% Compareing the original and compressed matrices visually
figure
subplot(1,2,1)
imagesc(x); axis image; colorbar;
title('Original Matrix')
xlabel('Column'); ylabel('Row');
subplot(1,2,2)
imagesc(y); axis image; colorbar;
title('Compressed Matrix')
xlabel('Column'); ylabel('Row');
```

```
     2     1
     3     2
     1     3
     1     4
     3     4

Energy ratio: 0.948640

Y2 =

     0     0     4     7
     8     0     0     0
     0   -13     0     4
```

```
K2 =

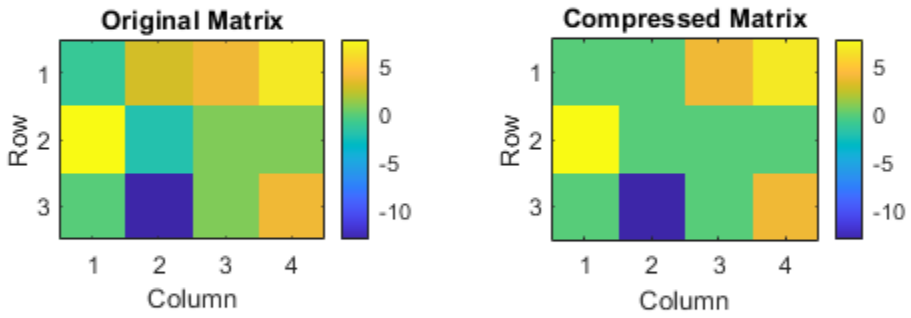
     2
     3
     1
     1
     3


L2 =

     1
     2
     3
     4
     4


R2 =

    0.9486
```




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